CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 442 Practice

12/01/2023 · 1.5 hours · 25 marks

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01 About this midterm

The midterm of semester 442, dated 12/01/2023, by the same instructor. The questions are copied from photos of a graded student paper.

Question	1	2	3	4	5	6	Total
Marks	2	4	6	8	3	2	25

Important

This midterm follows a different syllabus. It has heaps, binary search trees and graphs, which are not in this semester's lectures that I received. The parts useful to you are Q2 (true/false), Q3b and Q5 (hash tables), and Q6 (counting operations).

Two questions are not copied in full because their figures are unreadable in the photos: the graph of Q1 and the tree of Q4c. I have no official answer key, so the solutions are my own.

02 Questions

Q1 BIPARTITE GRAPH · 2 M

Check if this graph is bipartite or not, and why?

(The graph figure is not readable in the photo.)

The answer: _____ The reason is: _____

Q2 FILL IN + TRUE / FALSE · 2.5 + 1.5 M

a) Fill in the blanks

• InsertItem to dictionary takes _____ time.

• Bottom-up heap construction runs in _____ time.

• In the binary search tree: Let u, v, and w be three nodes such that u is in the left subtree of v and w is in the right subtree of v. Therefore we have: key(u) ≤ _____ ≤ key(w).

• The space used by a Queue with a Singly Linked list is _____

• FindElement in binary tree takes _____

b) Put T or F for each of the following statements.

( ) In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.

( ) Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.

( ) The total of primitive operations represents the Worst case of an algorithm.

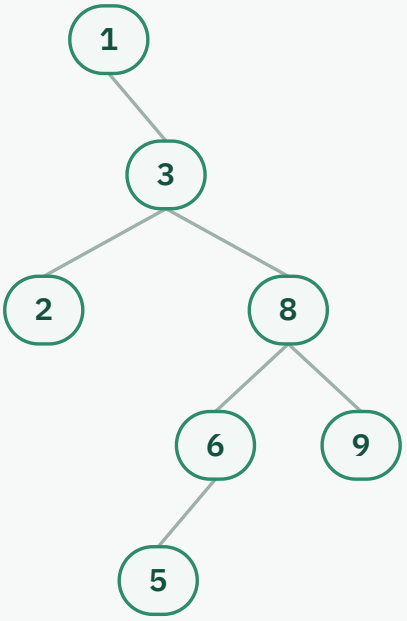
Q3 HEAPS + HASH FUNCTION · 4 + 2 M

a) what is the objective of Upheap, Downheap algorithms? when to use? then what time required to run UpHeap and DownHeap?

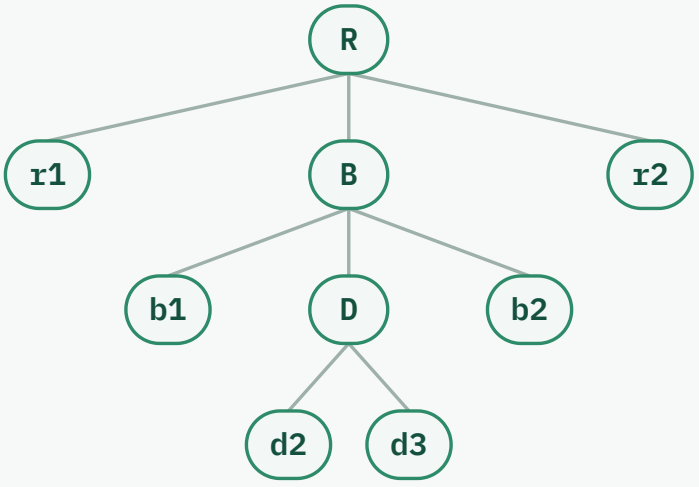
b) When you can say that the hash function is "good"?

Q4 TREES · 2 + 4 + 2 M

a) Remove the node 3 from the following binary tree, then redraw the tree after deletion.



b) Design the Linked Data Structure that represents the following Tree structure.



c) Write the sequence of tree traversal of the figure. *(The figure is cut off in the photo.)*

Q5 COLLISION HANDLING · 3 M

Discuss all available methods for handling collisions in hash tables.

Q6 COUNTING OPERATIONS · 2 M

Count the total primitive operations of the following algorithm.

```
Algorithm arrayMax(A, n)
   currentMax ← A[0]
   Temp ← A[0]
   for i ← 1 to n − 1 do
      if A[i] > currentMax then
         currentMax ← A[i]
   { increment counter i }
   return currentMax
```